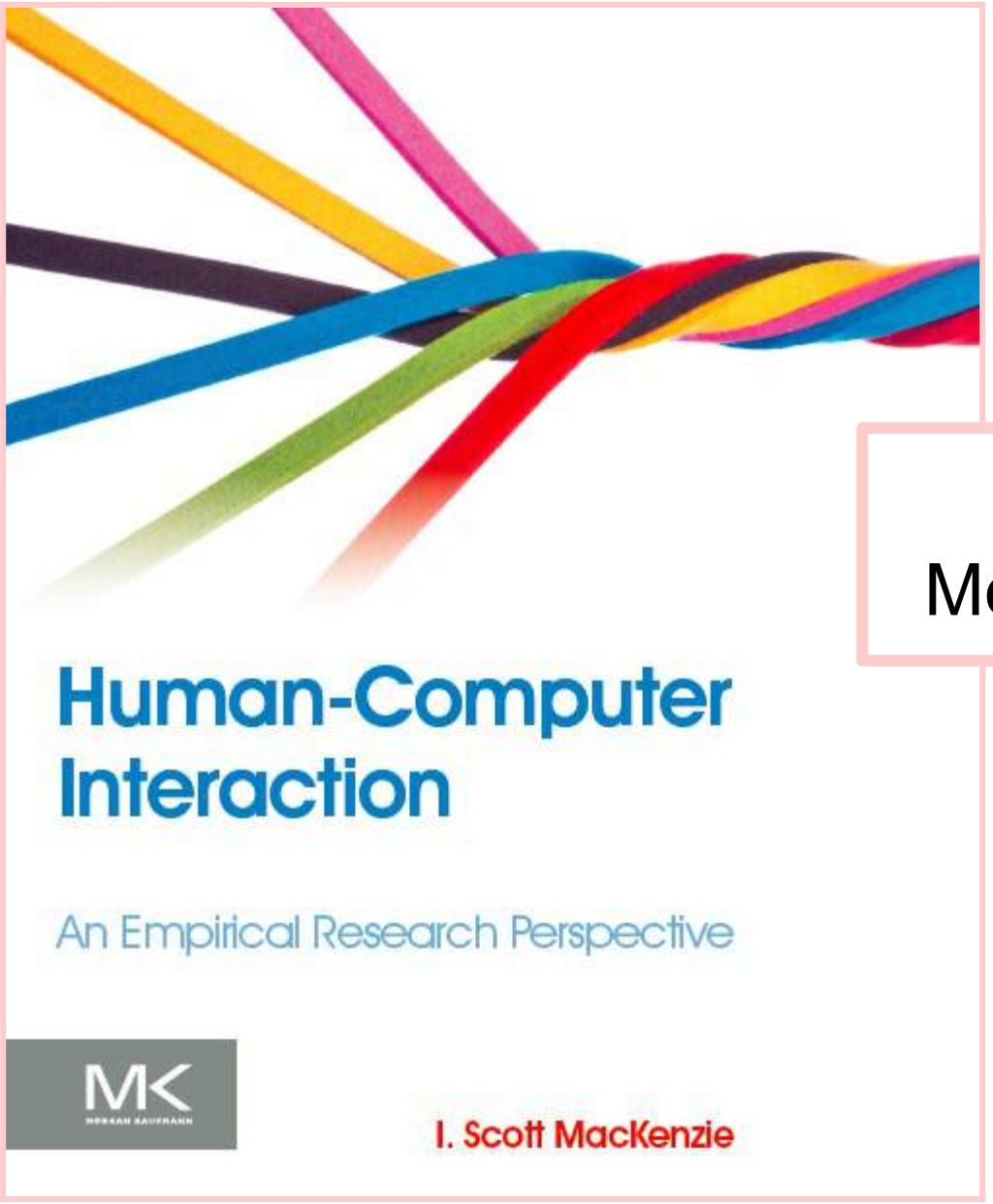


Chapter 7
Modeling Interaction

Longbiao CHEN
2018/04 Xiamen University

ASSIGNMENT 2

- Hand-in: **Drop-in-box @ B108E**
- Due: **15h 17/08/2018**
- Exam: **15-17h 17/08/2018**





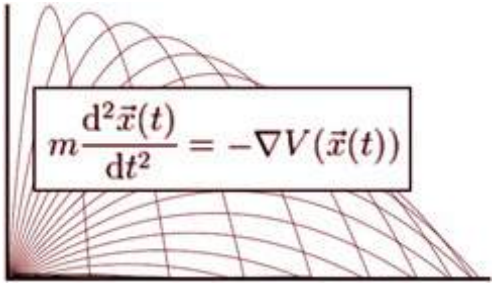
Introduction

- A **model** is a simplification of reality

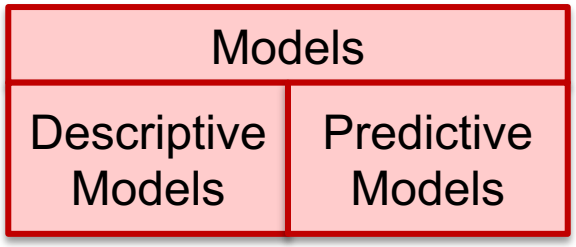
Architect's scale model of a building



Physicist's model for the trajectory of a tossed ball



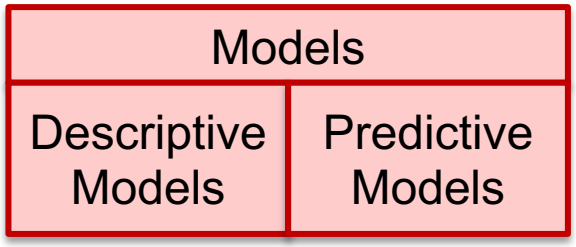
- Both are **simplifications** of complex phenomena
- The architect's model is a **description**
  - space usage, movement of people, light, shade, etc.
- The physicist's model is a **prediction**
  - the ball's position as a function of time



Descriptive Models

Descriptive Models

- Descriptive models are everywhere
- Descriptive modeling is at times so simple, the process barely seems like modeling
- Any **reduction** or **partitioning** of a **problem** space qualifies as a descriptive model, e.g.,



- Other names:
  - *Design space*, *framework*, *taxonomy*, *classification*, and often without a name given

Descriptive Model Examples

- Politics
- Groupware
- Keyboards
- Two-handed input
- Graphical input

Big Fuzzy Cloud Model of Politics

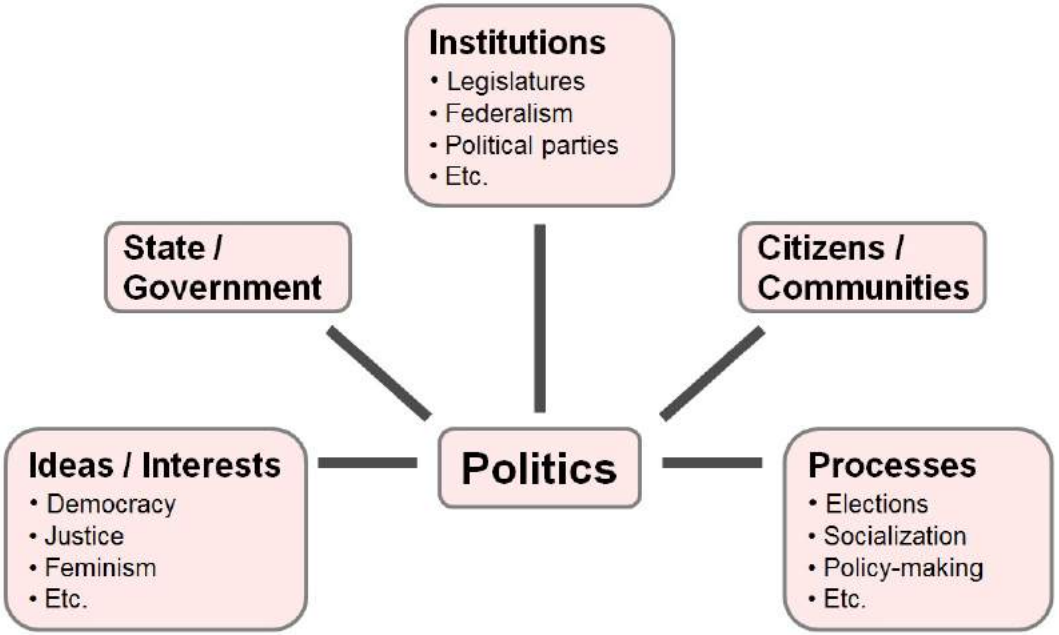
- Let's consider a topic we all know a little about
- Below is a *big fuzzy cloud* model for politics:



- Of course, there is no model – it's just the thing itself
- How do we go about making a descriptive model of politics?
- Break it down; partition the topic into parts
- What are the things that make up politics?
- How can they be labeled, presented, and organized?

Descriptive Model of Politics[1]

- With Johnston's model of politics, we are empowered to think differently about politics



Questions:

- *Is the model correct?*
- *Is there a different organization that might work better?*
- *Is it correct to mirror citizens/communities with state/government?*
- *Are the five perimeter items sufficient?*
- *Is federalism an institution (or an idea)?*
- *Is the model useful?*

Most important question

[1] Johnston, L. (2007*). Politics: An introduction to the modern democratic state* (3rd ed.). Peterborough, Ontario: Broadview Press.

Descriptive Model Examples

- Politics
- Groupware
- Keyboards
- Two-handed input
- Graphical input

CSCW (Groupware)

- A research topic within HCI is CSCW [computer supported cooperative work]
- Concerned with people **working collaboratively** using computing technology
- Aka *groupware*
- Same challenge:
  - How do we go about making a descriptive model of groupware?
  - Break it down; partition the topic into parts
  - What are the things that make up groupware?
  - How can they be labeled, presented, and organized?



Quadrant Model of Groupware[1]

- A descriptive model
- Groupware partitioned into a 2 × 2 space
  - Location → same place | different places
  - Time → same time | different times

	Same Time	Different Times
Same Place	Copy boards PC Projectors Facilitation Services Group Decision Room Polling Systems	Shared Files Shift Work Kiosks Team Rooms Group Displays
Different Places	Conference Calls Graphics and Audio Screen Sharing Video Teleconferencing Spontaneous Meetings	Group Writing Computer Conferencing Conversational Structuring Forms Management Group Voice Mail

[1] Johansen, R. (1991). Groupware: Future directions and wild cards. *Journal of Organizational Computing and Electronic Commerce*, *1*(2), 219-227.

Critiquing the Model

- The quadrant model of groupware was introduced in 1991
- The same questions apply:
  - *Is the model correct? Is there a different organization that might work better? Is the model useful? Etc.*
- Many of today's methods of collaborating didn't exist in 1991
- Contemporary groupware activities include
  - Sharing photos using camera phones, web cams, Skype, social media, blogging, tweeting
- Can these be positioned in the quadrant model of groupware? (probably; give it a try)

Descriptive Model Examples

- Politics
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- Graphical input

Keyboards

- Keyboards date at least to the 1870s with the introduction of the Scholes-Glidden typewriter keyboard
- Today's keyboards retain the same core letter arrangement (qwerty), but have many extra keys
- 100+ keys can produce a wide variety of letters, symbols, commands, etc.
- Same challenge:
  - How do we go about making a descriptive model of keyboards?
  - Break it down; partition the topic into parts
  - What are the things that make up keyboards?
  - How can they be labeled, presented, and organized?



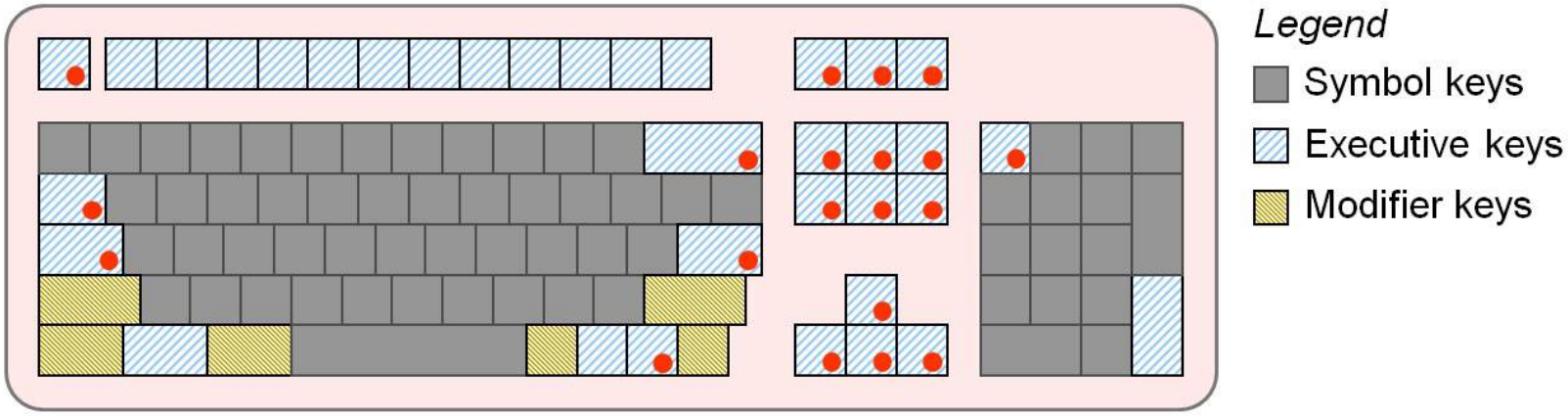
1874



2012

Key-Action Model (KAM)[1]

- A descriptive model
- Three categories of keys:
  - Symbol keys → produce graphic symbols (e.g., A)
  - Executive keys → invoke actions (e.g., ESC)
  - Modifier keys → modify effect of other keys (e.g., CTRL)



[1] MacKenzie, I. S. (2003). Motor behaviour models for human computer interaction. In J. M. Carroll (Ed.), *HCI models, theories, and frameworks: Toward a multidisciplinary science* (pp. 27-54): San Francisco: Morgan Kaufmann.

Critiquing the Model

- Nice visualization
  - Reveals organization of the keyboard in terms of symbol, executive, and modifier keys
- Questions:
  - *Is the model correct? Do all keys fit the model? Are there additional categories to improve the model? Do some keys fit more than one category? Can the model be applied to other keyboards, such as mobile phone keyboards or soft keyboards? Is the model useful? Etc.*
- Note: Red dots on previous slide identify executive keys that are not mirrored (excluding function keys)

Descriptive Model Examples

- Politics
- Groupware
- Keyboards
- Two-handed input
- Graphical input

Two-handed Input

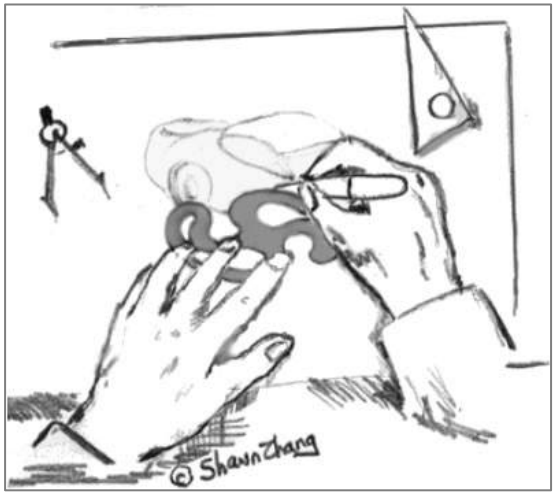
- Humans not only have two hands, they use their hands differently
- Most people have a hand preference (*Which hand do you use to write?*)
- Study of hand usage is called [bimanual control]



- Guiard undertook such a study, examining the roles of the **preferred and non-preferred hands** in common tasks
- The result is a descriptive model

Guiard's Model of Bimanual Control[1]

Hand	Role and Action
Non-preferred	• Leads the preferred hand • Sets the spatial frame of reference for the preferred hand • Performs coarse movements
Preferred	• Follows the non-preferred hand • Works within established frame of reference set by the non-preferred hand • Performs fine movements



Consider the bulleted points above in terms of the sketch on the right.

[1] Guiard, Y. (1987). Asymmetric division of labor in human skilled bimanual action: The kinematic chain as a model. *Journal of Motor Behavior, 19*, 486-517.

Scrolling

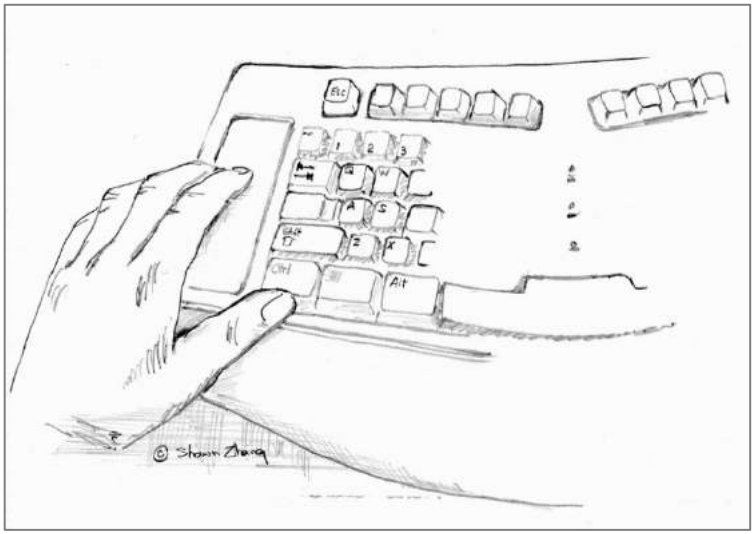
- The table below deconstructs scrolling as it relates to other common computing tasks that coexist with scrolling

Task	Characteristics
Scrolling	• Precedes/overlaps other tasks • Sets the frame of reference • Minimal precision needed (coarse)
Selecting, editing, reading, drawing, etc.	• Follows/overlaps scrolling • Works within frame of reference set by scrolling • Demands precision (fine)

- Do you see the similarity with Guiard's model of bimanual control?
- What does this suggest? (next slide)

Non-preferred Hand Scrolling

- Guiard's model of bimanual control suggests that scrolling is a task well suited to the non-preferred hand:



- But, what about the wheel mouse, introduced in 1996 by Microsoft as the *Intellimouse*?
- How to get scrolling into the non-preferred hand?

Re-engineered *Intellimouse*

- A presentation rationalizing non-preferred hand scrolling (in view of Guiard's model of bimanual control) was given at Microsoft in Feb 1998 using a re-engineered *Intellimouse*:



- In fact, the presentation was given twice the same day (a.m. → "mouse group" | p.m. → "keyboard group")
- The case for scrolling via the non-preferred hand was persuasive

Two-years later

Microsoft Office Keyboard

- Includes left-side wheel (roller) for scrolling via the non-preferred hand (for right-handed users):



- The left-side scroll wheel was eventually discontinued in later releases of Microsoft keyboards
- Momentum and popularity of the wheel mouse were too much to overcome

Descriptive Model Examples

- Politics
- Groupware
- Keyboards
- Two-handed input
- Graphical input



Graphical Input

- Considerable research on GUIs followed the successful introduction of the Apple *Macintosh* in 1984
  - Common interactive techniques (tasks):
    - pointing, dragging, selecting, inking, rubber-banding, texting
  - Common technologies (devices):
    - mouse, trackball, touch panel, joystick, stylus, finger

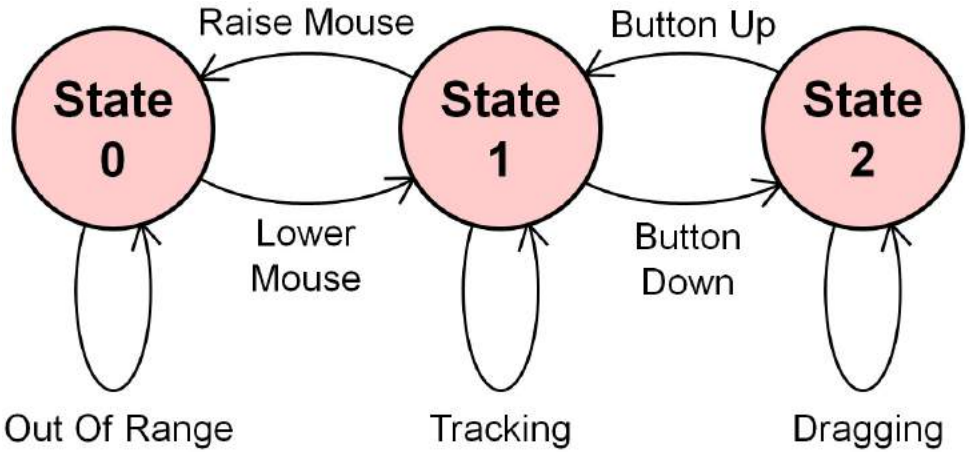
- How can the tasks and devices be reconciled and understood to promote better designs?

- **A three-state model of graphical input**

[1] Buxton, W. (1990). A three-state model of graphical input. *Proceedings of INTERACT '90*, 449-456, Amsterdam: Elsevier.

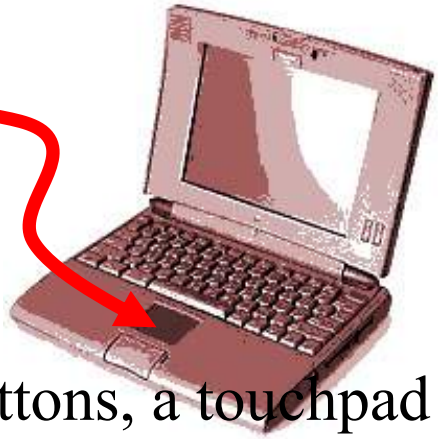
Buxton's Three State Model of Graphical Input

- A descriptive model
- Expresses GUI interaction in terms of three states
- Mouse example: (different for other devices)
  - State 0 → out of range
  - State 1 → tracking
  - State 2 → dragging



[Wikipedia: Buxton's Three-State-Model]

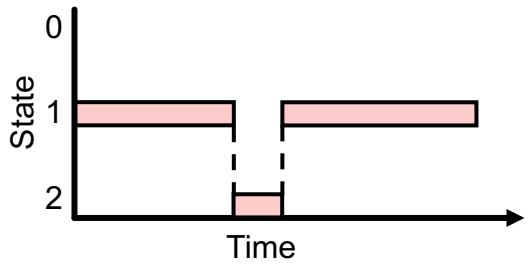
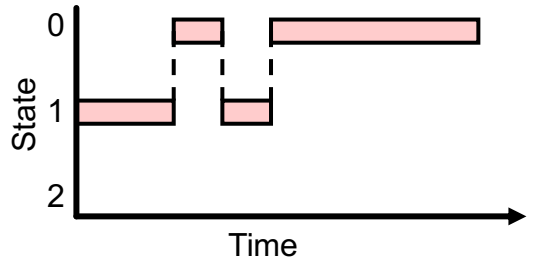
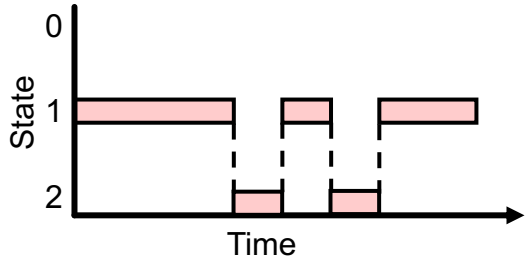
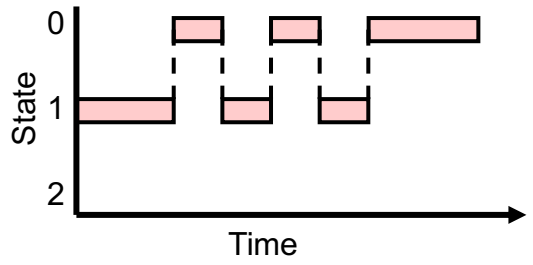
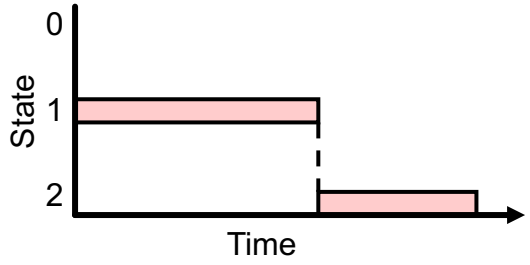
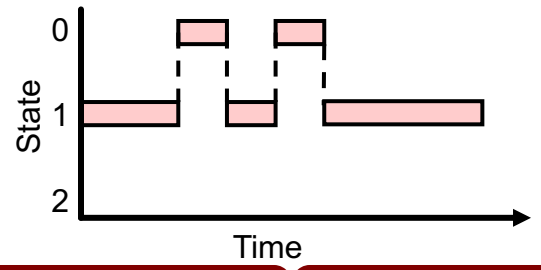
Application of Buxton's Model



- In 1994 Apple introduced the *Trackpad* on its [Powerbook 500] notebook computer
- Trackpad (Touchpad) became the standard pointing device on notebook computers
- Besides physical buttons to mimic mouse buttons, a touchpad includes "*lift and tap*" to implement button down/up actions using touch
- But, [lift-and-tap] actions are error prone → during a tap, if the finger moves before lifting, a *dragging* action is sometimes invoked, instead of a click

Analysis

- Let’s analyse and compare common touchpad and mouse interactions, guided by Buxton’s three-state model (next slide)

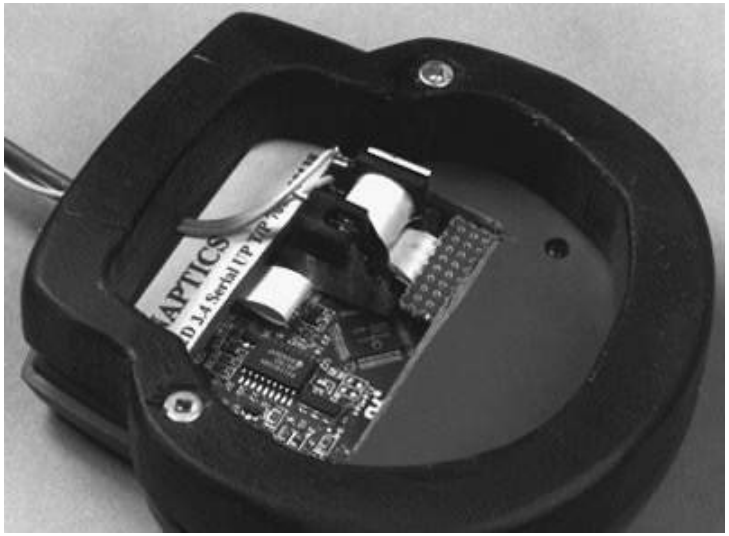
	Mouse	Touchpad
Click		
Double Click		
Drag		

Extra State Transitions

The Tactile Touchpad[1]

- Uses native *x*-*y*-*z* mode (*z* = finger pressure)
- Implements button down (state 2) by "pressing harder"
- Button click feedback provided by relay below touchpad
- Design guided by Buxton's three-state model





[1] MacKenzie, I. S., & Oniszczak, A. (1997). The tactile touchpad. *Extended Abstracts of the ACM SIGCHI Conference on Human Factors in Computing Systems - CHI '97*, 309-310, New York: ACM.

Demo



[http://www.youtube.com/watch?v=fxfu-Yo6yEk]

Three-state Touch Products

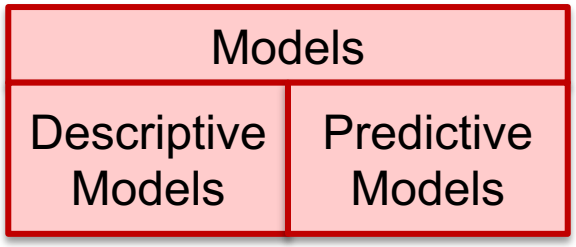


[Blackberry *Storm* (2008)]



Apple *Macbook* (2008)

[The new MacBook (2015)]



Predictive Models

Predictive Models

- A predictive model is an **equation**
- Predicts the outcome on a *criterion variable* (aka *dependent variable* or *human response*) based on the value of one or more *predictor variables* (aka *independent variables*)
- However, predictive models deal with **numbers**, not concepts

Why Use Predictive Models

- Card et al. presented perhaps the first predictive model in HCI.[1] In many respects, their work was straight-forward experimental research; but they went further:
  - "While these empirical results are of direct use in selecting a pointing device, it would obviously be of greater benefit if a theoretical account of the results could be made. For one thing, the need for some experiments might be obviated; for another, ways of improving pointing performance might be suggested."
- This is a call for the use of predictive models in HCI
- They went on to present predictive models using **Fitts' law**

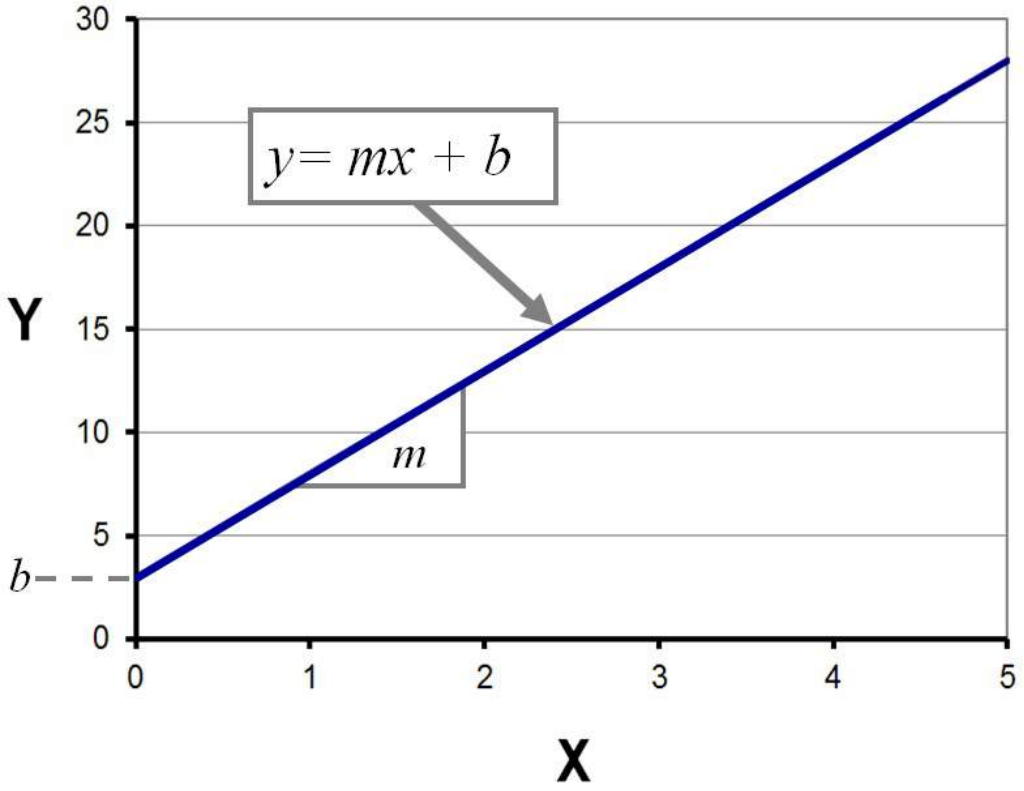
[1] Card, S. K., English, W. K., & Burr, B. J. (1978). Evaluation of mouse, rate-controlled isometric joystick, step keys, and text keys for text selection on a CRT. *Ergonomics, 21*, 601-613.

Predictive Model Examples

- Linear prediction equation
- Fitts' law
- Choice reaction time
- Keystroke-level model (KLM)
- Skill acquisition
- More than one predictor

Linear Prediction Equation

- The basic prediction equation expresses a linear relationship between a predictor variable (x) and a criterion variable (y):



Predictive Model Examples

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Fitts' Law

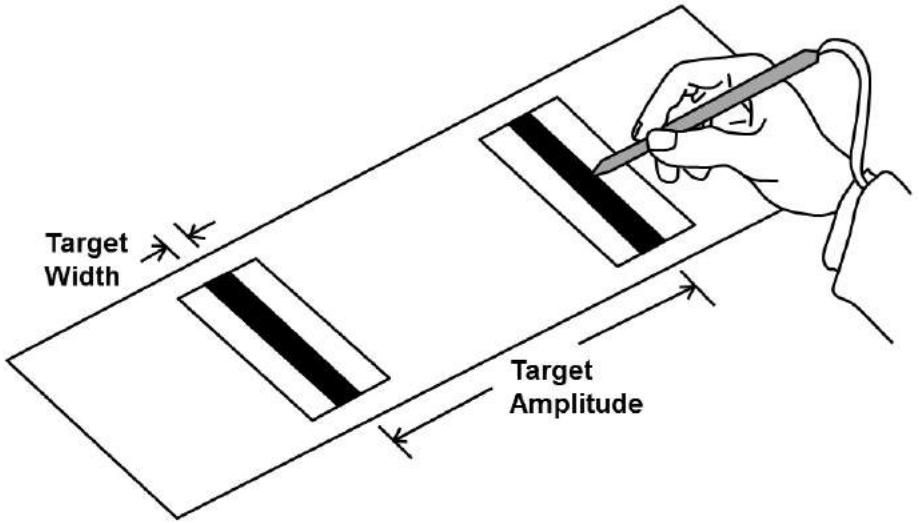
- One of the most widely used models in HCI
- Model for rapid aimed movements (e.g., moving a cursor toward a target and selecting the target)
- Three applications:
  1. Use a Fitts' law prediction equation to analyse and compare design alternatives
  2. Use Fitts' *index of performance* (now *throughput*) as a dependent variable in a comparative evaluation
  3. Determine if a device or technique "conforms to Fitts' law"

Predictive model

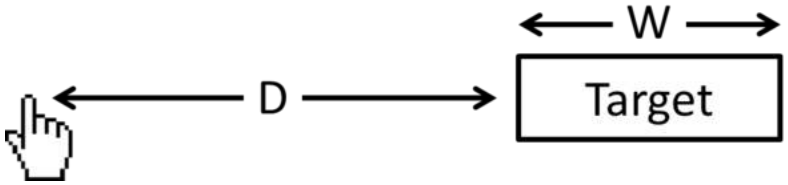
- Origins: Two highly-cited papers in experimental psychology, one from 1954[1], the other for 1964[2]

[1] Fitts, P. M. (1954). The information capacity of the human motor system in controlling the amplitude of movement. *Journal of Experimental Psychology, 47*, 381-391.
[2] Fitts, P. M., & Peterson, J. R. (1964). Information capacity of discrete motor responses. *Journal of Experimental Psychology, 67*, 103-112.

Fitts' Law – Task Paradigms



$$T = a + b \log_2 \left(1 + \frac{D}{W} \right)$$



These sketches were adapted from Fitts' 1954 and 1964 papers. It is easy to imagine comparable tasks implemented on computing technology.

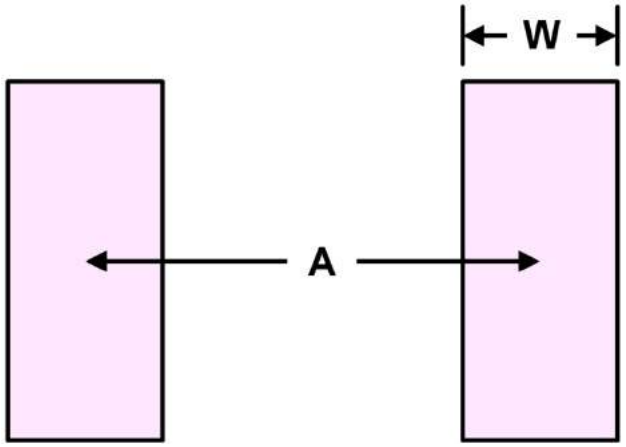
[Fitts' Law Experiment]

Fitts' Index of Difficulty (*ID*)

- Fitts' index of difficulty (*ID*) is a measure of the difficulty of a target selection task:

$$ID = \log_2\left(\frac{A}{W} + 1\right)$$

Units: bits



- Normally the prediction equation is built using the *effective* index of difficulty (ID_e) – includes an "adjustment for accuracy"
- Fitts hypothesized that the relationship between movement time (*MT*) and *ID* is linear

Fitts' Law Models for Pointing Devices

- A research project compared four pointing devices, including two for remote pointing[1]
- Twelve participants performed a series of serial target selection tasks using the four devices
- For our purpose, we'll look at the data and models for two of the devices:



Interlink *RemotePoint*



Microsoft *Mouse 2.0*

[1] MacKenzie, I. S., & Jusoh, S. (2001). An evaluation of two input devices for remote pointing. *Proceedings - EHCI 2000*, 235-249, Heidelberg, Germany: Springer-Verlag.

Experiment Conditions and Observations

Conditions

Mouse Observations

RemotePoint Observations

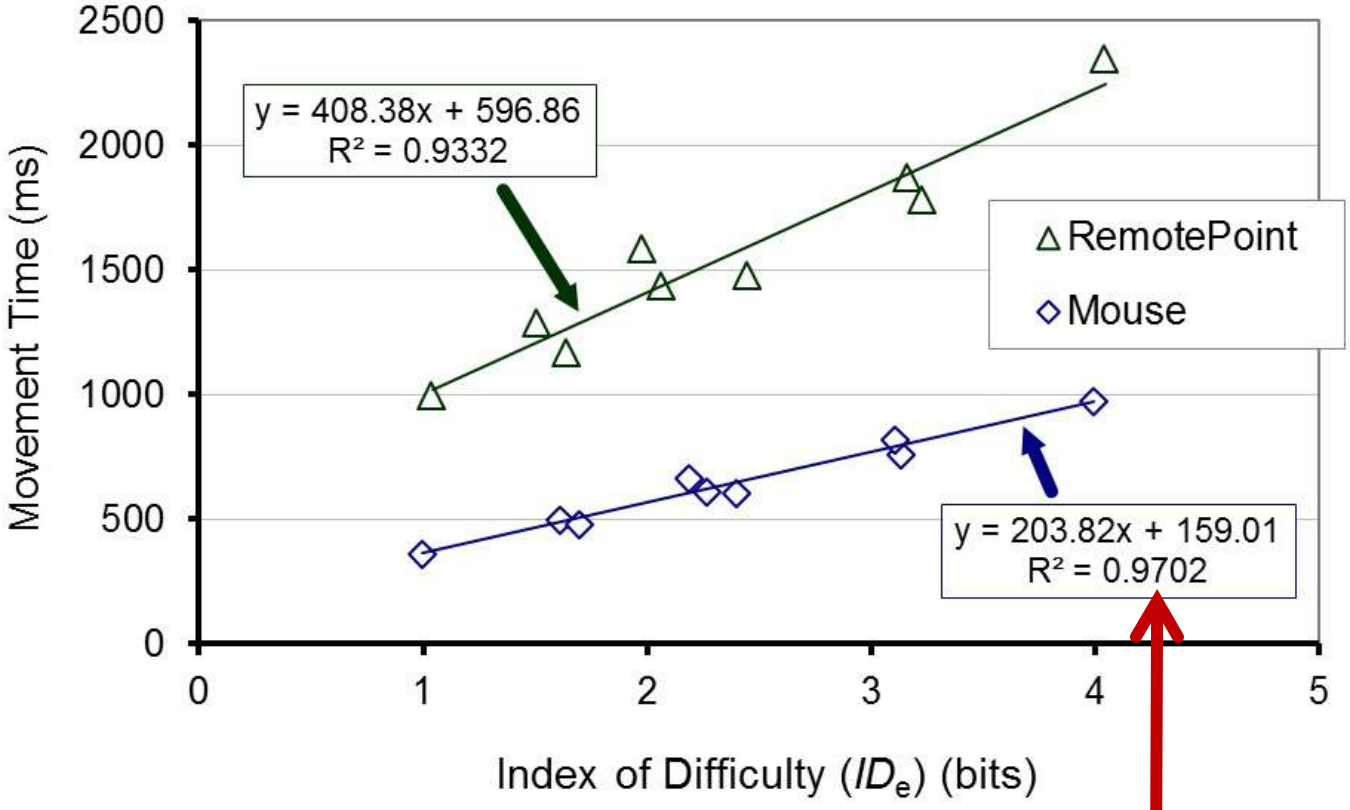
A (pixels)	W (pixels)	ID (bits)	Mouse				RemotePoint			
			W_e (pixels)	ID_e (bits)	MT (ms)	TP (bits/s)	W_e (pixels)	ID_e (bits)	MT (ms)	TP (bits/s)
40	10	2.32	11.23	2.19	665	3.29	13.59	1.98	1587	1.25
40	20	1.58	19.46	1.61	501	3.21	21.66	1.51	1293	1.17
40	40	1.00	40.20	1.00	361	2.76	37.92	1.04	1001	1.04
80	10	3.17	10.28	3.13	762	4.11	10.08	3.16	1874	1.69
80	20	2.32	18.72	2.40	604	3.97	25.21	2.06	1442	1.43
80	40	1.58	35.67	1.70	481	3.53	37.75	1.64	1175	1.40
160	10	4.09	10.71	3.99	979	4.08	10.33	4.04	2353	1.72
160	20	3.17	21.04	3.11	823	3.77	19.09	3.23	1788	1.81
160	40	2.32	41.96	2.27	615	3.69	35.97	2.45	1480	1.65
		Mean	23.25	2.38	644	3.60	23.51	2.35	1555	1.46

For model building…

x sample points
y sample points

x sample points
y sample points

Fitts' Law Prediction Equations



Squared correlations are very high.
Yes, the *MT-ID* relationship is linear!

Calculation of Throughput (*TP*)

- Two approaches in the literature
  - 1. $TP = ID / MT$
  - 2. $TP = 1 / b$ (where b is the slope of the regression line)
- Are they the same?
- Mouse example just presented
  - Method 1
    - $TP = 2.38 / 0.644 = 3.70$ bits/s (but see MacKenzie, 2015)[1]
  - Method 2
    - $TP = 1 / 0.204 = 4.90$ bits/s
- Method 1 is used when using *TP* as a dependent variable
- Method 2 cannot be used because the influence of the intercept is absent

[1] MacKenzie, I. S. (2015). Fitts' throughput and the remarkable case of touch-based target selection. *Proc HCII 2015*, pp. 238-249. Berlin: Springer.

Predictive Model Examples

- Linear prediction equation
- Fitts' law
- Choice reaction time
- Keystroke-level model (KLM)
- Skill acquisition
- More than one predictor

Choice Reaction Time

- Given *n* stimuli, associated one-for-one with *n* responses, the time to react to the onset of a stimulus is the *choice reaction time*
- Modeled by the Hick-Hyman law:[1] [2]

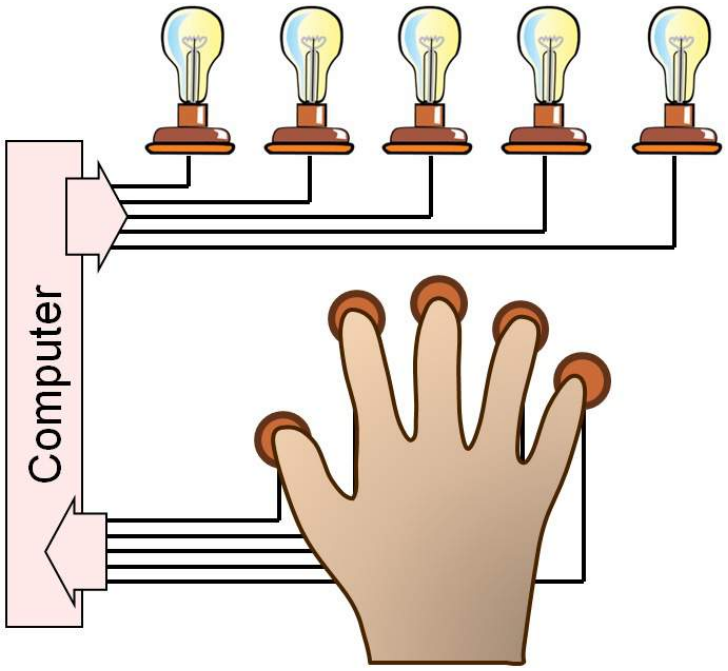
$$RT = a + b \log_2(n + 1)$$

- Coefficients:

  $a \approx 200$ ms

  $b \approx 150$ ms/bit

- An Information processing model (like Fitts' law)



[1] Hick, W. E. (1952). On the rate of gain of information. *Quarterly J Exp Psychol, 4*, 11-36.

[2] Hyman, R. (1953).Stimulus information as a determinant of reaction time. *J Exp Psychol, 45*, 188-196.

HCI Applications

- Not many, but…
  - A **[telephone operator]** selects among ten buttons when a light behind a button turns on[1]
  - Time to select items in a hierarchical menu (visual search eliminated by practicing participants to expert levels)[2]
  - Activation time for mode switching with non-dominant hand in a tablet interface[3]
- Difficult to apply because additional behaviours are often present, such as visual search or movement

[1] Card, S. K., Moran, T. P., & Newell, A. (1983). *The psychology of human-computer interaction*. Hillsdale, NJ: Erlbaum.

[2] Landauer, T. K., & Nachbar, D. W. (1985). Selection from alphabetic and numeric menu trees using a touch screen: Breadth, depth, and width. *Proc CHI '85*, 73-77, ACM.

[3] Ruiz, J., Bunt, A., & Lank, E. (2008). A model of non-preferred hand mode switching. *Proceedings of Graphics Interface 2008*, 49-56, Toronto: Canadian Information Processing Society.

Predictive Model Examples

- Linear prediction equation
- Fitts' law
- Choice reaction time
- Keystroke-level model (KLM)
- Skill acquisition
- More than one predictor

Keystroke-Level Model (KLM)[1] [2]

- One of the earliest and most comprehensive models in HCI
- Developed specifically for predicting human performance with interactive computing systems
- Predicts expert error-free task completion times
- Elements of a KLM prediction
  - Task (or a series of tasks)
  - Method used
  - Command language of the system
  - Motor skill parameters of the user
  - Response time parameters of the system

[Keystroke Level-Model Demonstration]

[1] Card, S. K., Moran, T. P., & Newell, A. (1980, July). The keystroke-level model for user performance time with interactive systems. *Communications of the ACM, 23*, 396-410.
[2] Card, S. K., Moran, T. P., & Newell, A. (1983). *The psychology of human-computer interaction*. Hillsdale, NJ: Erlbaum.

Why Use the KLM?

- Consider a task such as “delete a file”
- Perhaps there are two ways to do the task:
  1. Mouse + menu selection
  2. Keyboard + command entry
- The KLM can predict the time for each method
- If used at the design stage, design alternatives may be considered and compared → design choices follow

A KLM Prediction

- A task is broken into a series of subtasks
- Total predicted time is the sum of the subtask times:

$$t_{EXECUTE} = t_K + t_P + t_H + t_D + t_M + t_R$$

- Operators:
  - K → keystroking P → pointing H → homing
  - D → drawing M → mental prep R → system response
- Some operators are omitted or repeated, depending on the task (e.g., if n keystroking operations are required, t_K becomes $n \times t_K$)
- Operator values (next slide)

KLM Operators and Values

Operator	Description	Time (s)
K	PRESS A KEY OR BUTTON Pressing a modifier key (e.g., shift) counts as a separate operation, Time varies with typing skill: Best typist (135 wpm) Good typist (90 wpm) Average skilled typist (55 wpm) Average non-secretary typist (40 wpm) Typing random letters Typing complex codes Worst typist (unfamiliar with keyboard)	 0.08 0.12 0.20 0.28 0.50 0.75 1.20
P	POINT WITH A MOUSE Empirical value based on Fitts' law. Range from 0.8 to 1.5 seconds. Operator does *not* include the button click at the end of a pointing operation	1.10
H	HOME HAND(S) ON KEYBOARD OR OTHER DEVICE	0.40
D(n_D.l_D)	DRAW n_D STRAIGHT-LINE SEGMENTS OF TOTAL LENGTH l_D. Drawing with the mouse constrained to a grid.	.9 n_D + .16 l_D
M	MENTALLY PREPARE	1.35
R(t)	RESPONSE BY SYSTEM Different commands require different response times. Counted only if the user must wait.	t

Original KLM Experiment

- The KLM was validated in an experiment with fourteen tasks performed using various methods and systems
- Example: Task 1 → *Replace a 5-letter word with another word (one line from previous task)*
- Using one system, POET, the task was broken down as follows:

Jump to next line	**M K**[LINEFEED]
Issue Substitute command	**M K**[S]
Type new word	**K**[word]
Terminate new word	**M K**[RETURN]
Type old word	**K**[word]
Terminate old word	**M K**[RETURN]
Terminate command	**K**[RETURN]

- 4 mental operations + 15 keystroking operations, hence →

KLM Prediction (Example)

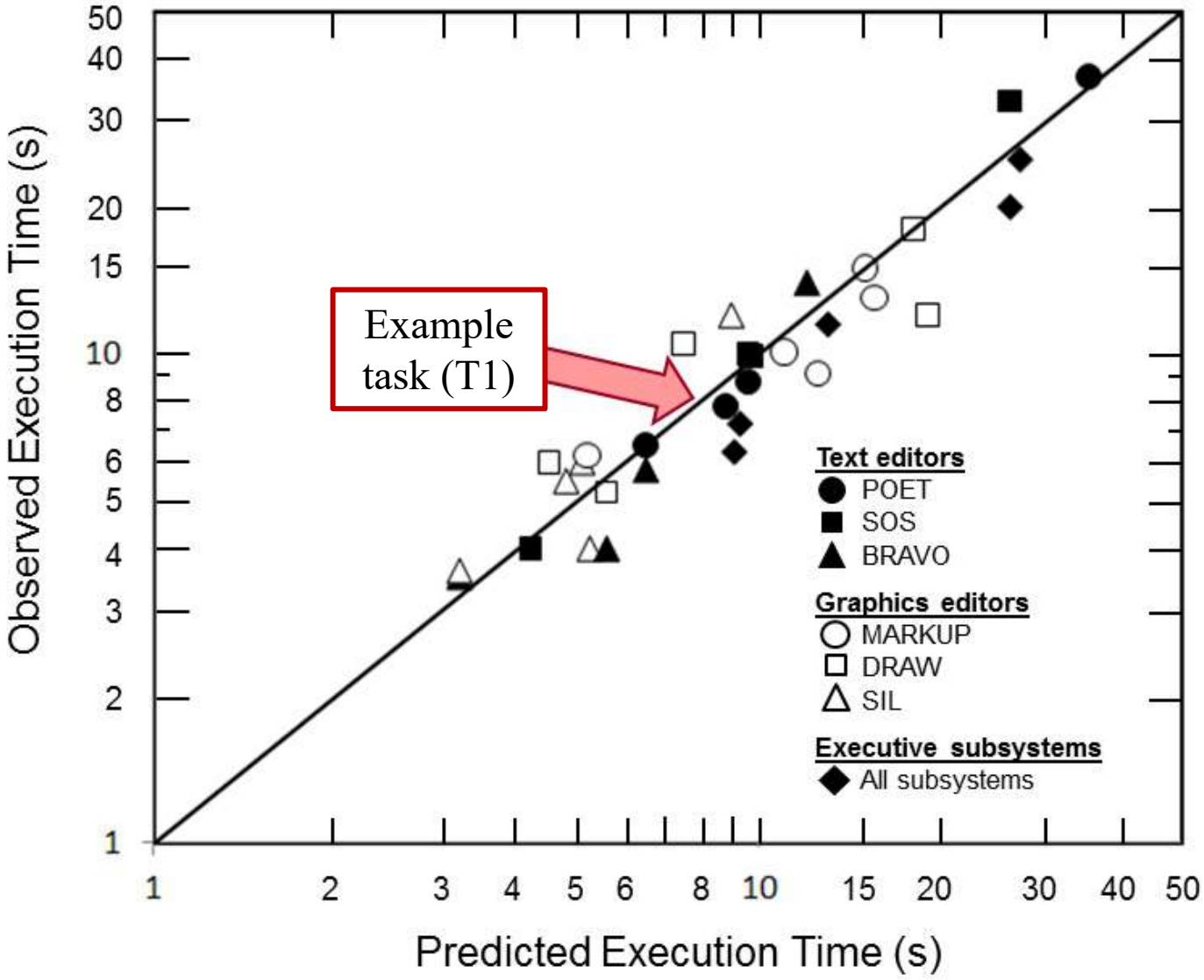
$$t_{\text{EXECUTE}} = 4 \times t_{\text{M}} + 15 \times t_{\text{K}}$$

- M set to 1.35 seconds (two slides back)
- K set to 0.23 seconds, based on a 5-minute pre-test
- So…

$$t_{\text{EXECUTE}} = 4 \times 1.35 + 15 \times 0.23$$
$$= 8.85 \text{ seconds}$$

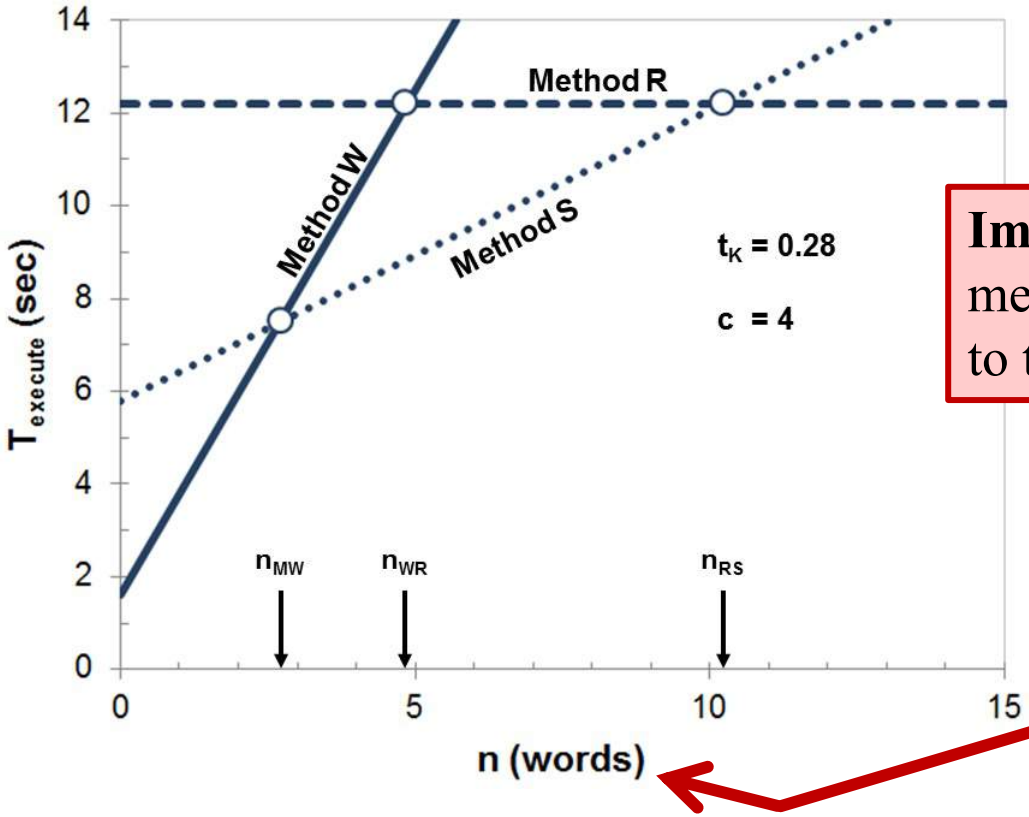
- This is the prediction
- What about the observation? (next slide)

Results


Observed Execution Time (s)
Predicted Execution Time (s)
50
40
30
20
15
10
8
6
5
4
3
2
1
1
2
3
4
5
6
8
10
15
20
30
40
50
Example task (T1)
Text editors
POET
SOS
BRAVO
Graphics editors
MARKUP
DRAW
SIL
Executive subsystems
All subsystems

Sensitivity Analysis

- If parameters are treated as variables, the sensitivity of predictions to changes in parameters can be assessed



Implication: The preferred method changes with the distance to the misspelled word.

n is a parameter – the distance in words to the location of a misspelled word to correct.

Predictive Model Examples

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Skill Acquisition

- When learning a skill, we begin as *novices*
- Initial performance is poor, but, with practice, we acquire skill
- With continued practice, we become proficient, perhaps *experts*
- The novice to expert transition is well suited to predictive modeling
- Dependent variable → proficiency (typically, the time or speed in doing a task)
- Independent variable → amount of practice (e.g., hours, days, months, blocks, sessions)

Power Law of Learning

- Relationship between proficiency and practice is non-linear:
  - At first, a small amount of practice yields substantial improvement
  - Later, the same "small amount of practice" yields only a slight improvement
- Relationship best expressed by a power function:

$$y = b \times x^{a}$$

(general form)

$$T_n = T_1 \times n^{a}$$

(power law of learning)

where…

T_n → time to do the task on the nth trial

T_1 → time to do the task on the 1st trial (a constant)

n → trial indicator (e.g., hours, days, blocks, sessions)

a → a constant setting the shape of the curve

Note: *a* is negative since task completion time *decreases* with practice

Speed Variation

- Dependent variable can be speed (*S*), the reciprocal of time
- Model predicts "tasks per unit time" (e.g., words per minute)
- Mathematical form:

$$S_n = S_1 \times n^a$$

where…

S_n → speed on the nth trial

S_1 → speed on the 1st trial (a constant)

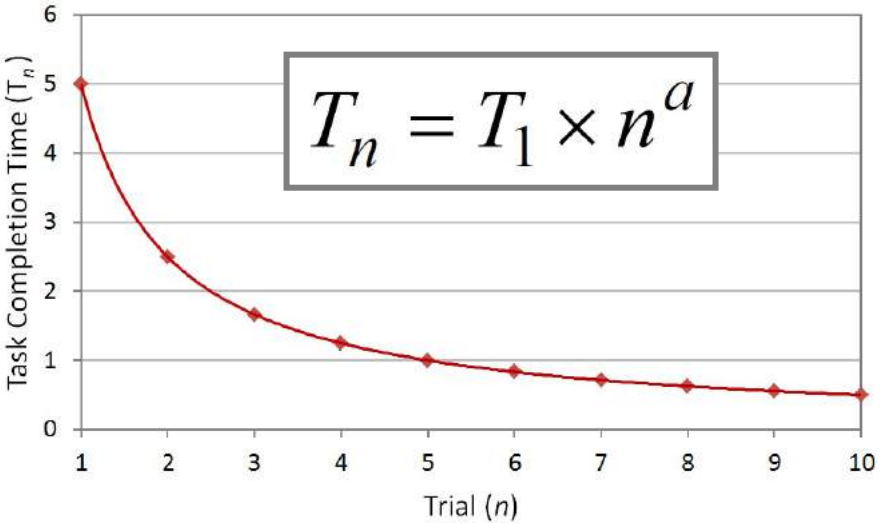
n → trial indicator (e.g., hours, days, blocks, sessions)

a → a constant setting the shape of the curve

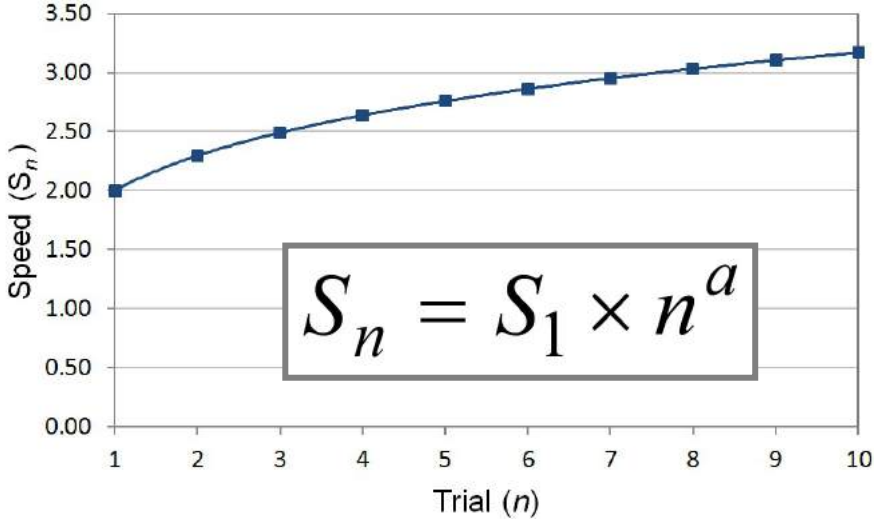
Note: a is positive and <1 reflecting the diminishing return with practice

Curve Shapes

Predicting time



Predicting speed



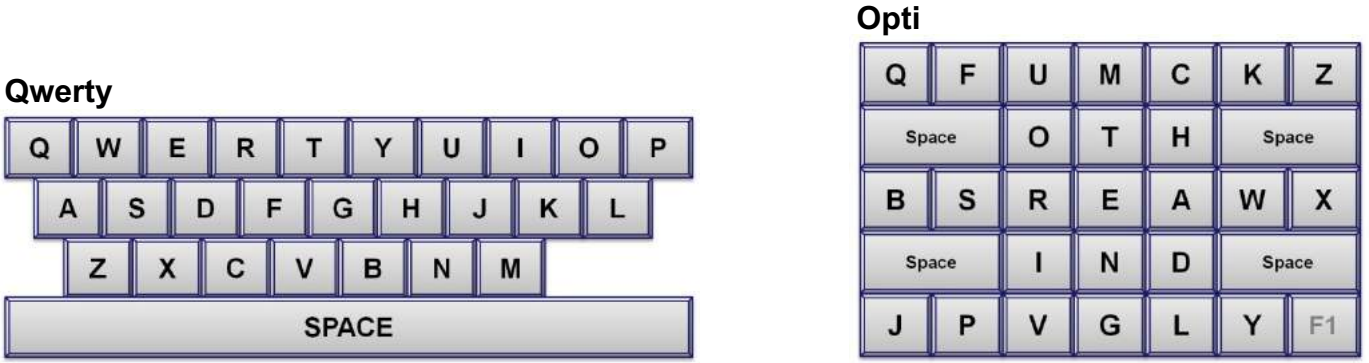
Demo

[click here] to open Excel spreadsheet

(PowerLawOfLearning.xlsx)

Example

- An experiment compared two soft keyboards for text entry[1]
  - Qwerty → conventional letter arrangement
  - Opti → optimized to minimize finger or stylus movement



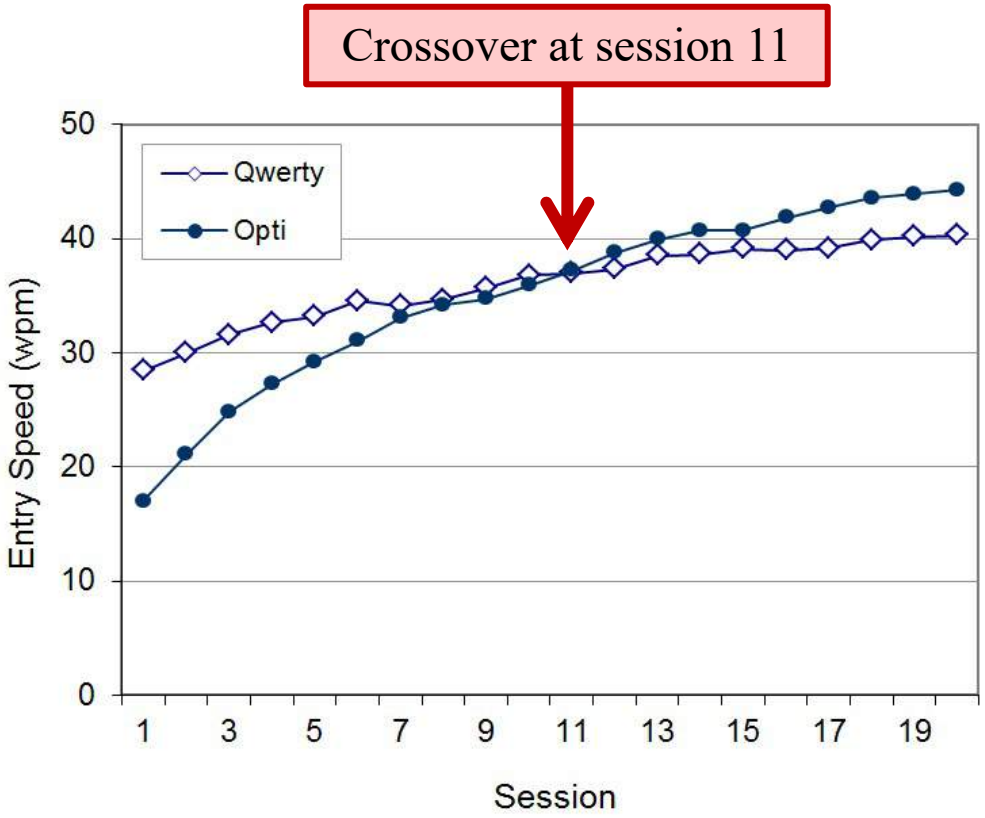
- Qwerty expected to be faster initially
- Opti expected to be faster with practice
- Participants performed 20 sessions of text entry; results →

[1] MacKenzie, I. S., & Zhang, S. X. (1999). The design and evaluation of a high-performance soft keyboard. *Proc CHI '99*, 25-31, New York: ACM.

Results

Entry Speed (wpm)

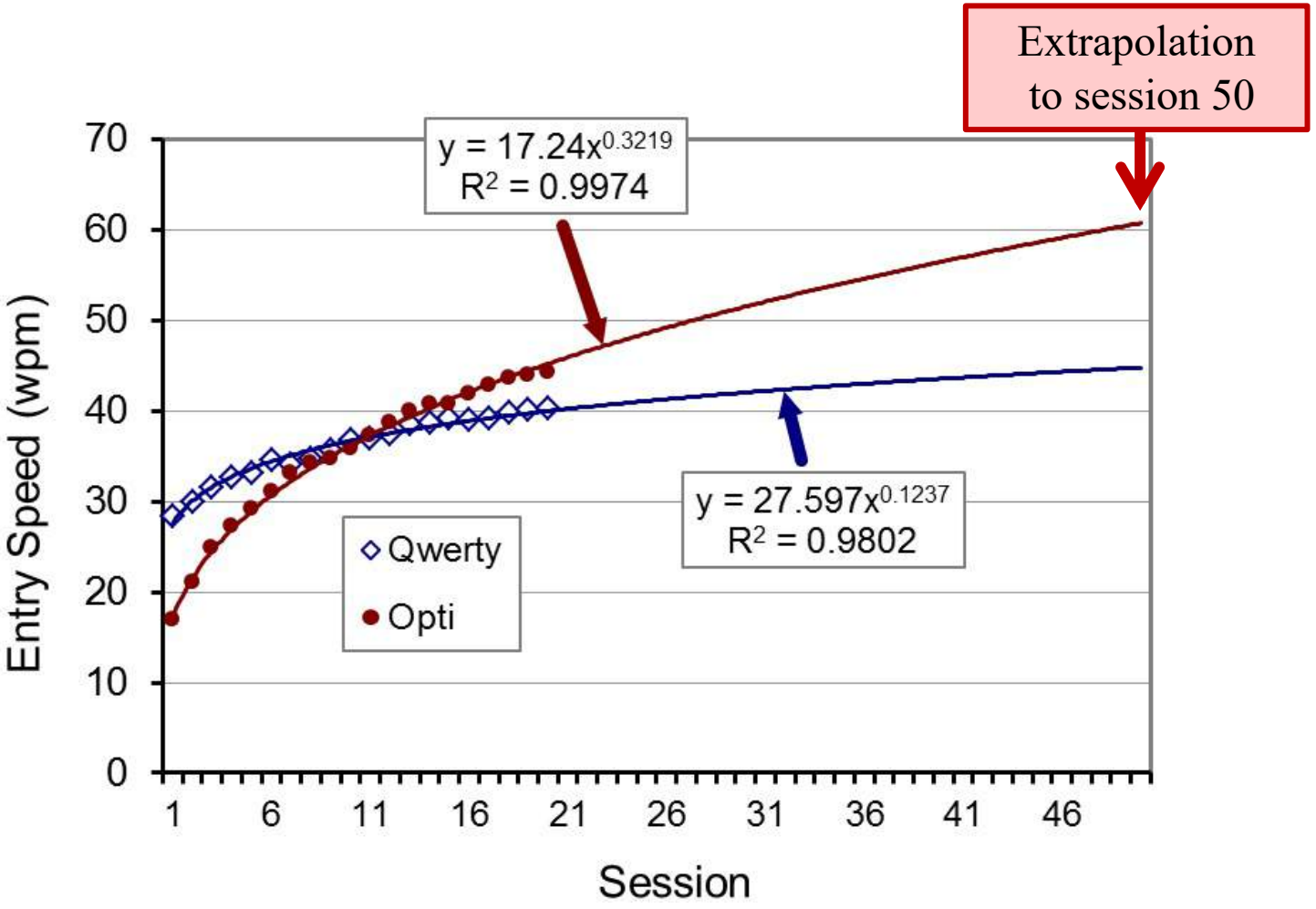
Session	Qwerty	Opti
1	28.44	16.98
2	29.98	21.06
3	31.56	24.80
4	32.62	27.26
5	33.20	29.18
6	34.52	31.04
7	34.14	33.08
8	34.68	34.16
9	35.66	34.72
10	36.77	35.89
11	36.95	37.22
12	37.36	38.75
13	38.50	39.95
14	38.65	40.73
15	39.12	40.68
16	39.00	41.85
17	39.20	42.74
18	39.85	43.55
19	40.16	43.91
20	40.30	44.29



Power law of learning?

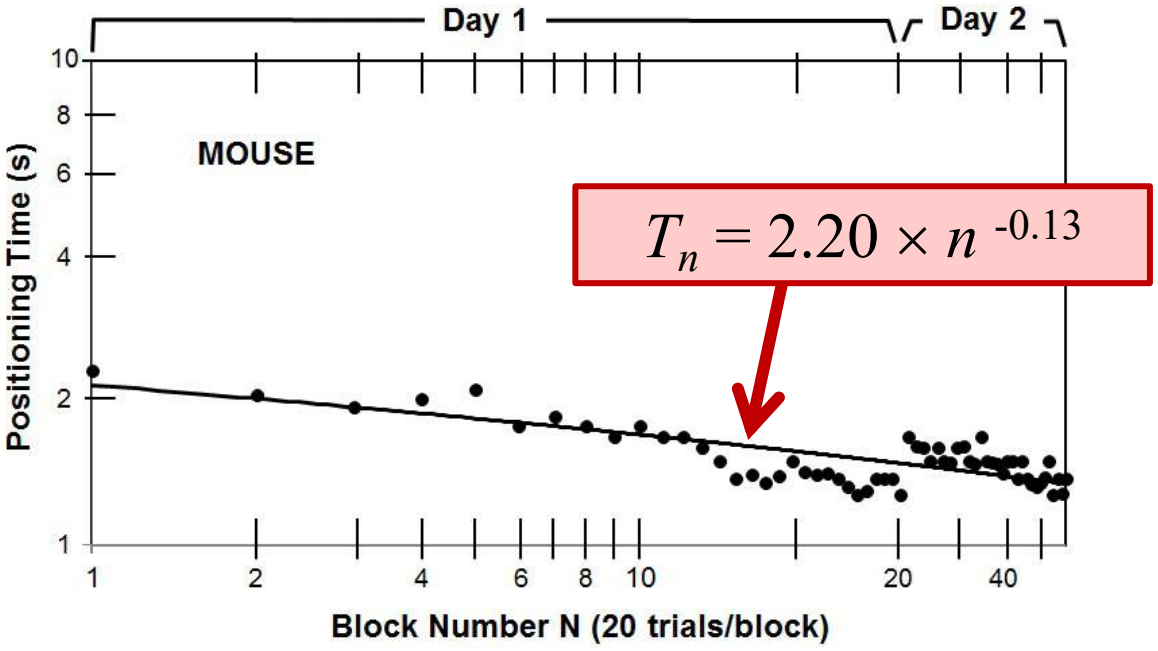
(next slide)

Power Law of Learning



Log-log Model

- If the *x* and *y* data are transformed to log scales, the relationship is linear
- Example:[1]



[1] Card, S. K., English, W. K., & Burr, B. J. (1978). Evaluation of mouse, rate-controlled isometric joystick, step keys, and text keys for text selection on a CRT. *Ergonomics, 21*, 601-613.

Predictive Model Examples

- Linear prediction equation
- Fitts' law
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- More than one predictor

More Than One Predictor

- A prediction equation can have more than one predictor:

$$y = a + b_1 x_1 + b_2 x_2 + b_3 x_3 + \ldots$$

- In statistics, the technique is called *multiple regression*

Hypothetical Example

- Research question:
  - Is there a relationship between the time to learn a computer game and the age and computing habits of players?
- Dependent variable:
  - y → the time to reach a criterion score
- Predictor variables:
  - x_1 → age of player
  - x_2 → daily computer use in hours
- The experiment involved 14 participants (next slide)

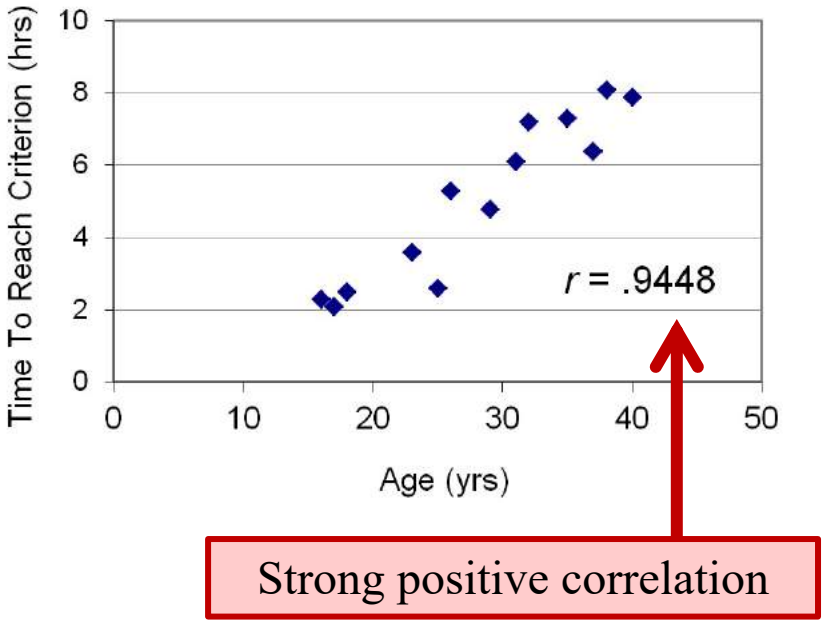
Data

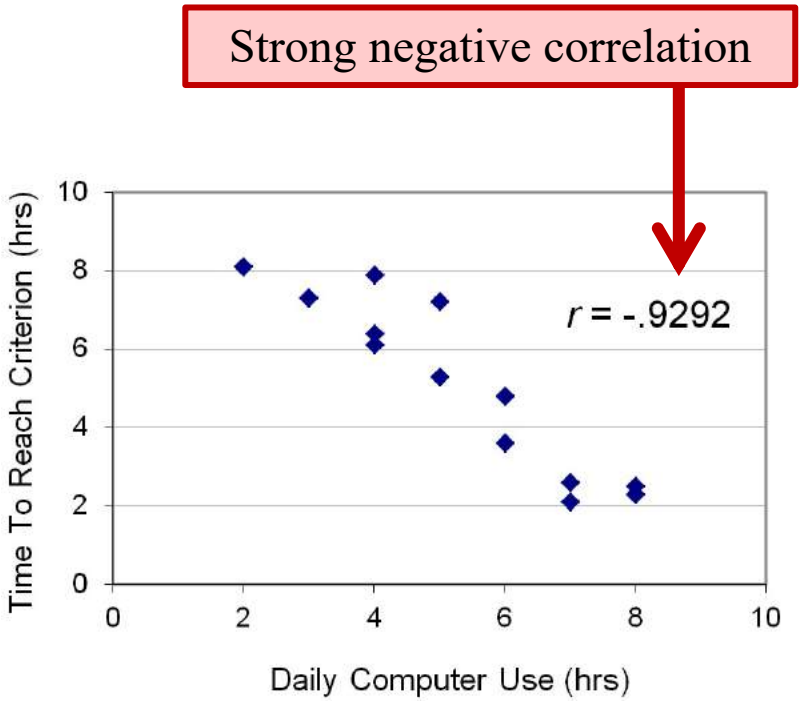
Participant	Time To Reach Criterion (hours)	Age (years)	Daily Computer Usage (hours)
P1	2.3	16	8
P2	2.1	17	7
P3	2.5	18	8
P4	3.6	23	6
P5	2.6	25	7
P6	5.3	26	5
P7	4.8	29	6
P8	6.1	31	4
P9	7.2	32	5
P10	7.3	35	3
P11	6.4	37	4
P12	8.1	38	2
P13	7.9	40	4
P14	2.3	16	8

Determined experimentally

Determined in pre-test interviews

Scatter Plots and Correlations



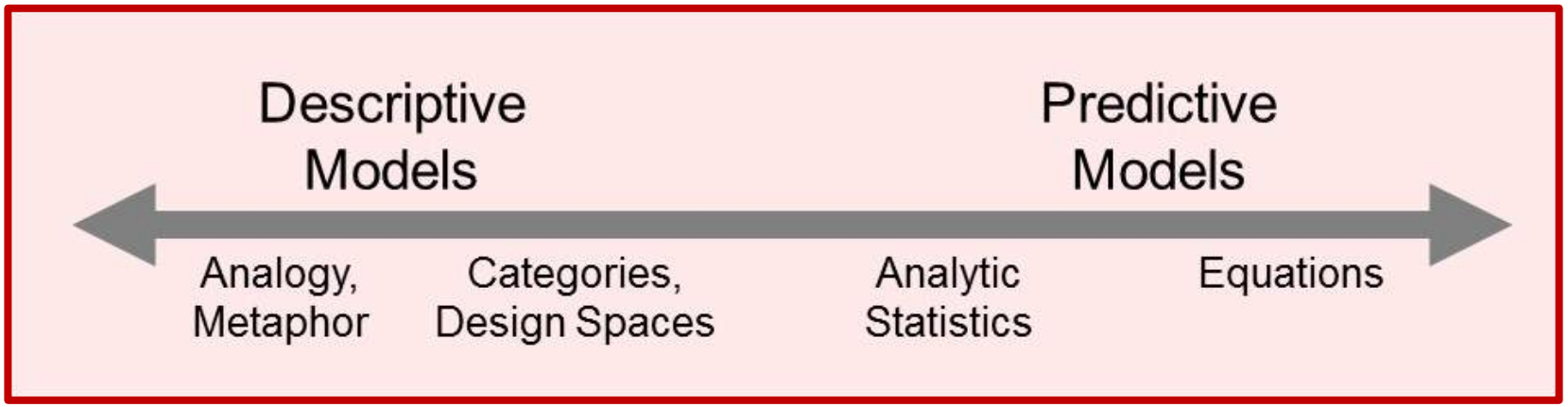


Prediction equation?

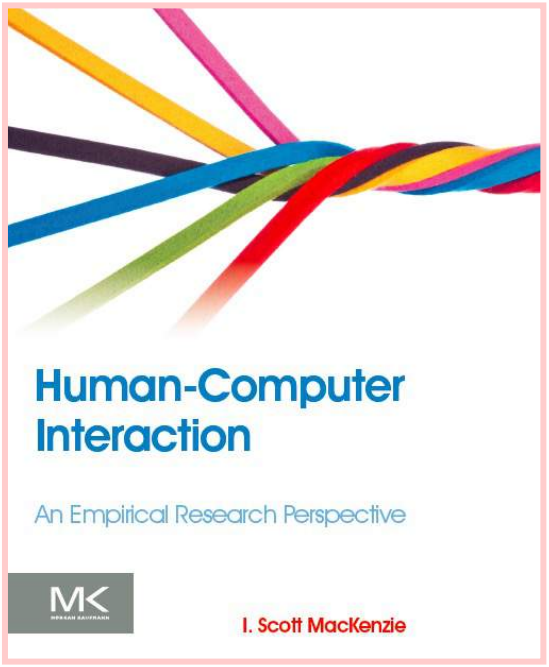
(next slide)

Post Script

- Models presented here were of two types:
  - Descriptive models
  - Predictive models
- The "space" for modeling is likely richer and more diverse
- Below is a Model Continuum Model (MCM)



Thank You


Human-Computer Interaction
An Empirical Research Perspective
MK
I. Scott MacKenzie